Chapter 27

Statistics

FUNDAMENTAL MEASUREMENT ASSUMPTIONS

The four basic assumptions on which the validity of all measurements depends are *random distribution*, *fixed model*, *fixed variation*, and *fixed distribution*. The majority of statistical tests depend very heavily on the assumption that the data were collected in a random manner. This is an important assumption because of the significant probability during measurement of the occurrence of aggregations, changes in condition, assignable causes, trends, cyclical effects, and progressive learning, and drift in natural, industrial, and human processes. The random distribution assumption is particularly important when calculating statistics that are the basis of an inference or comparison; for example, taking a sample that will be the basis for making an inference as to the population it is drawn from. Another example of the random distribution assumption is in experiments, where the results of one treatment are compared to those of another factor setting.

A commonly used definition for normality of data distributions is IID (independently and identically distributed), which is defined in NIST SP 800-90B as "[a] quality of a sequence of random variables for which each element of the sequence has the same probability distribution as the other values, and all values are mutually independent."[1] This assumption is often made regarding data that have been randomly selected for use in statistical calculations. This is in a Normal distribution with mean zero (0) and variance (s^2).

DEGREES OF FREEDOM

Degrees of freedom (often represented by *n* or its Greek letter equivalent n, nu), in relation to the count of data items on which a statistic is based, refers to the number of ways a statistic's value can vary with a variation in each of the source data items. It is one of the first items determined in the calculation of many statistics. When the number of data items is very large or includes the total population of items on which an inference is being made, the symbol *N* is often used. Conversely, when the degrees of freedom (DoF) relate to a statistic based on a sample from a larger population that is being described, the symbols *n* or n are most often used. This is to help identify the meaning to the user.

Because determining DoF involves summation of the number of data items being evaluated (minus a constant in some cases), the resulting DoF, being a count, is dimensionless (not associated with a unit). The determination of DoF depends

on the net ways that a statistic may vary with variation in the data elements going into it. For instance, in the calculation of the mean statistic based on 20 values, the resulting statistic can vary 20 different ways with the change of any of the source data values. Hence, the DoF for mean is merely the count, *N*, of the data items.

In contrast, as mentioned, some statistics have a DoF that is the count of data items used in the statistic calculation minus a constant. For instance, in the calculation of the sample standard deviation, part of the calculation involves first calculating the mean. Because the mean statistic is part of the sample standard deviation calculation, one degree of freedom is lost due to the use of the mean, and the resulting DoF is the data count minus one.

This is not a book on statistical theory, so the formulas for various statistics will be given and their DoF basis merely listed without discussion as to the mathematical basis for them.

$$\mathrm{df} = N - 1$$

RESIDUALS

A *residual* is the difference between the value of a data item and a statistic describing it. For instance, if the average of a set of measurements is *x*-bar = 4.583, and the first two values are $x_1 = 4.282$ and $x_2 = 4.632$, the residuals of these values is the difference between the measurement value and the average, or $r_1 = (4.282–4.583) = –0.301$ and $r_2 = (4.632–4.583) = 0.049$.

Because the operation involved in calculating residuals is subtraction, the resulting statistic has the same units as the data from which it is calculated. For instance, if measurements of voltage are used to calculate the mean and residuals for some or all of the data, the units for the residuals are volts.

CENTRAL TENDENCY

This is obviously the case because quite often the objective is to understand where the central value is. In mathematical terms, such statistics are said to determine central tendency. The primary statistics for describing central tendency are the mean, median, and mode.

Mean

One of the most common and often used or referred-to statistics is the *mean* or *average*. Because the operations involved in calculating the mean are addition and division, the resulting statistic has the same units as the data from which it is calculated. The mean or average is given by the following equation:

$$\overline{x} = \frac{\sum_{i-1}^{n} x_i}{n}$$

For example, in the data series of amperage measurements:

13, 14, 23, 23, 32, 33, 45, 99, and 105 A,

the mean (or average) value is 44.375 A.

Use Excel's AVERAGE() function. For example, for the data named range DATA, use the Excel function expression "=AVERAGE(DATA)".

Median

The *median* statistic is the middle value in a set of data items arranged in order of increasing value. That is, the median is the value where half the data items are less in value and half are greater in value. Increasingly, we are hearing this statistic being used for economic and demographic purposes, as in median income, median house value, median age.

To determine the median of a set of measurements, arrange the values in order of increasing value. If the number of values in the data set is odd, the median is the middle value in this ordered set. If the number of values in the data set is even, the median is the average of the middle two values.

For example, in the data series of amperage measurements:

13, 14, 23, 23, 32, 33, 45, 99, and 105 A,

the median (or middle) value is 32.

Use Excel's MEDIAN() function. For example, for the data named range DATA, use the Excel function expression "=MEDIAN(DATA)".

Mode

The *mode* statistic describes the most often (in statistics, described as most frequently) occurring value.

For example, in the data series:

13, 14, 23, 23, 32, 33, 45, 99, and 105 A,

the mode (or most often occurring) value is 23 A.

Use Excel's MODE() function. For example, for the data named range DATA, use the Excel function expression "=MODE(DATA)".

BIMODAL DISTRIBUTIONS

Some distributions either have more than one mode or have frequencies of occurrence of values where the histograms indicate more than one peak.

Such distributions are not normal distributions, that is, distributions that likely fit a normal distribution model, and in these cases statistics based on the assumption of a normal distribution may not be valid. Figure 27.1 is a sample graph of a bimodal distribution.

CENTRAL LIMIT THEOREM

The central limit theorem states that, for large enough sample size n, the distribution of the sample mean $\bar{x}$ will approach a normal distribution (see Figure 27.2). This is true for a sample of independent random variables from *any* population distribution, as long as the population has a finite standard deviation σ.[2]

3. If the original population is distributed normally (in other words, bell shaped), the sampling distribution of means will also be normal. If the original population is not normally distributed, the sampling distribution of means will increasingly approximate a normal distribution as sample size increases (in other words, when increasingly large samples are drawn).

CONFIDENCE INTERVAL

A *confidence interval* gives an estimated range of values, which is likely to include an unknown population parameter, the estimated range being calculated from a given set of sample data.

Confidence Intervals for Unknown Mean and Known Standard Deviation[4]

For a population with unknown mean μ and known standard deviation σ, a confidence interval for the population mean, based on a simple random sample (SRS) of size n, is

$$\overline{x} \pm z^* \, \sigma/\sqrt{n}$$

where z^* is the upper (1-C)/2 critical value for the standard normal distribution (see Figure 27.3).

Note: This interval is only exact when the population distribution is normal. For large samples from other population distributions, the interval is approximately correct by the central limit theorem.[5]

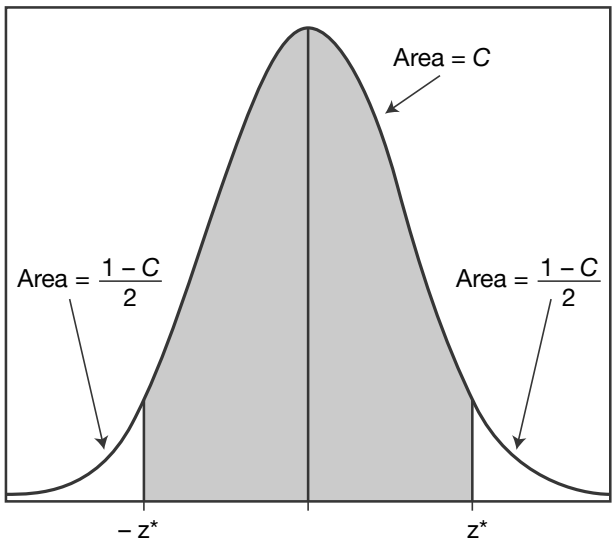


Figure 27.3 Confidence intervals.

ROOT MEAN SQUARE

The *root mean square* (RMS) statistic is most often associated with continuous variables such as time. There are occasions where RMS is associated with discrete variables as well, such as specific measurements of a variable, for example, weight. As the name implies, RMS involves squaring the values of data, finding the mean (average) of these values, followed by taking a (square) root of this mean. As such, the unit of RMS is the same as the measurement unit. For instance, if the RMS value of a set of data is taken in "meter," the RMS result also has the unit meter.

Root mean square (RMS) is given by the following equation:

$$x_{\text{rms}} = \sqrt{\frac{\sum_{i=1}^{n} \Upsilon^2}{n}}$$

For instance, in the data series of amperage measurements:

13, 14, 23, 23, 32, 33, 45, 99, and 105 A,

the RMS value is 54.15 A, rounded to two decimal places.

SUM OF SQUARES

Sum of squares (SS) is often used as one step in ANOVA calculations. The unit of SS is the measurement unit squared. The sum of squares is given by the following equation:

$$SS_{Total} = \sum (x_i - \overline{x})^2$$

SS_{Total} = Sum Squared Total Error

Σ = sum of data points

x_i = each data point

$\overline{x}$ = mean value

For instance, in the data series of amperage measurements:

13, 14, 23, 23, 32, 33, 45, 99, and 105 A,

the SS value is 26,387 A.

ROOT SUM OF SQUARES

The unit of *root sum of squares* (RSS) is the same as the measurement unit. The RSS is given by the following equation(s):

$$\text{RSS} = \sqrt{X_1^2 + X_2^2 + X_3^2 + \ldots + X_n^2}$$

For instance, in the data series of amperage measurements

13, 14, 23, 23, 32, 33, 45, 99, and 105 A

the RSS value is 162.44 A, rounded to two decimal places.

POPULATION VARIANCE

Unlike standard deviation, variances may be combined by addition. The **population variance**, *V*, is given by the following equation:

$$V = \sigma^2 = \frac{\sum_{i=1}^{n}(x_i - \mu)^2}{N}$$

σ^2 = population variance

x_i = value of the ith element

μ = population mean

N = population size

For example, in the data series of amperage measurements:

13, 14, 23, 23, 32, 33, 45, 99, and POP105 A,

the variance, *V*, value is 1201.23 A, rounded to two decimal places.

Use Microsoft Excel™ VARP() function. For example, for the data range named DATA, use the Excel™ function expression "=VARP(DATA)".

POPULATION STANDARD DEVIATION

The **population standard deviation** is used in the determination of several other statistics, for example, in determining a confidence interval and for hypothesis testing.

The population standard deviation, s, is given by the following equation:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \mu)^2}{N}} = \sqrt{\frac{\sum_{i=1}^{n}x_i^2}{N} - \mu}$$

For example, in the data series of amperage measurements:

13, 14, 23, 23, 32, 33, 45, 99, and 105 A,

the population standard deviation, s, value is 34.66 A, rounded to two decimal places.

Use Excel's STDEVP() function. For example, for the data range named DATA, use the Excel function expression "=STDEVP(DATA)".

SAMPLE VARIANCE

As noted, variances may be combined by addition. The **sample variance,** *v*, is given by the following equation:

$$v = s^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}$$

s^2 = sample variance

X_i = value of the ith element

$\bar{X}$ = sample mean

n = number of observations

For example, in the data series of amperage measurements

13, 14, 23, 23, 32, 33, 45, 99, and 105 A

the sample variance, *v*, value is 1372.84 A, rounded to two decimal places.

Use Excel's VAR() function. For example, for the data range named DATA, use the Excel function expression "=VAR(DATA)".

SAMPLE STANDARD DEVIATION

The **sample standard deviation** is used in the determination of several other statistics; for example, in determining a confidence interval and for hypothesis testing. The **sample standard deviation**, *s*, is given by the following equations:

$$s = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}} \quad s = \sqrt{\frac{\sum_{i=1}^{n}\left(X_i - \frac{\sum X_i}{n}\right)^2}{n-1}} \quad s = \sqrt{\frac{n\sum_{i=1}^{n}X_i^2 - \left(\sum_{i=1}^{n}X_i\right)^2}{n(n-1)}}$$

For example, in the data series of amperage measurements:

13, 14, 23, 23, 32, 33, 45, 99, and 105 A,

the sample standard deviation, *s*, value is 37.05 A, rounded to two decimal places

Use Excel's STDEV() function. For example, for the data range named DATA, use the Excel function expression "=STDEV(DATA)".

STANDARD ERROR OF THE MEAN

Standard error of the mean (SEM) does not assume a normal distribution. Many applications of SEM do assume a normal distribution. For SEM, the larger the sample size, the smaller the standard error of the mean. In other words, the size of SEM is inversely proportional to the square root of the sample size.

SEM, or *s*-*y*-bar, is given by the following formula:

$$s_{\bar{y}} = \frac{s}{\sqrt{n}}$$

where s = sample standard deviation and n = number of items in the sample. For example, in the data series of amperage measurements,

13, 14, 23, 23, 32, 33, 45, 99, and 105 A

the sample standard deviation, *s*, was 37.05 and the number of items, *n*, was 8, making the SEM 13.10, rounded to two decimal places.

MEASURES OF SKEWNESS AND KURTOSIS[6]

A fundamental task in many statistical analyses is to characterize the location and variability of a data set. A further characterization of the data includes skewness and kurtosis.

SKEWNESS

Skewness is a measure of symmetry (or lack of symmetry). Data sets generally are not completely normally distributed. There may be more low values than high values, resulting in histograms of the data that are higher on one side or the other of the mean (see Figure 27.4).

There are data sets where more values are below the mean value, and the graphed shape of the distribution leans toward higher, more positive values. Such distributions have a *positive skewness* (the value of the distribution's skew is positive, or greater than zero).

There are other data sets where more values are above (at a more positive value than) the mean, and the distribution appears to lean toward the more negative values. Such distributions have a *negative skewness* (the value of the distribution's skew is negative, or less than zero).

Distribution skewness is determined by the formula shown in Figure 27.4.[7] For example, in the data series of amperage measurements:

13, 14, 23, 23, 32, 33, 45, 99, and 105 A,

the skewness value is 1.15, dimensionless, rounded to two decimal places.

Use Excel's SKEW() function. For example, for the data range named DATA, use the Excel function expression "=SKEW(xx:yy)" where xx:yy is the cell location of your data (for example, B3:B18).

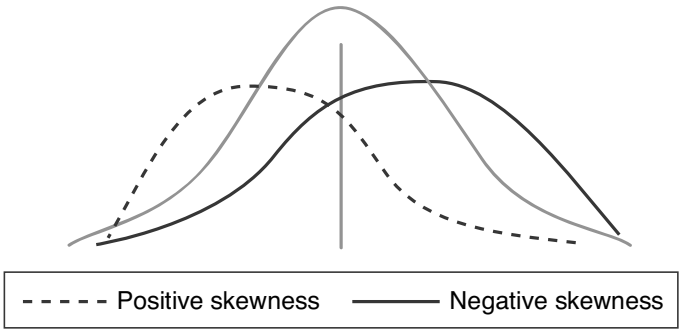


Figure 27.4 Example of positive and negative skewness.

KURTOSIS

Kurtosis is a measure of whether the data are heavy-tailed or light-tailed relative to a normal distribution. That is, data sets with high kurtosis tend to have heavy tails, or outliers. Data sets with low kurtosis tend to have light tails, or lack of

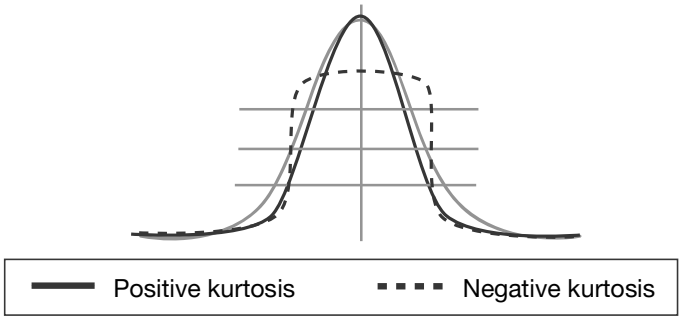


Figure 27.5 Example of positive and negative kurtosis.

outliers. A uniform distribution would be the extreme case.[8] Other data sets, not completely normally distributed, may have more values toward the tails than around the mean, and so on, resulting in histograms of the data that are more rectangular-shaped than bell curve (normal distribution)–shaped.

In the data sets where more values are toward the tails, and the distribution appears to be more rectangular or flat, the distribution is said to have negative kurtosis. Alternately, if more data are located closer to the mean, the distribution appears to be more triangular or peaked, and the distribution is said to have positive kurtosis (see Figure 27.5).

Distribution kurtosis is determined by the following formula:

$$\text{kurtosis} = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum_{i=1}^{n} \left(\frac{x_i - \overline{x}}{s} \right)^4 - \frac{3(n-1)^4}{(n-2)(n-3)}$$

For example, in the data series of amperage measurements

13, 14, 23, 23, 32, 33, 45, 99, and 105 A

the kurtosis value is –0.41, dimensionless, rounded to two decimal places.

Use Excel's KURT() function. For example, for the data range named DATA, use the Excel function expression "=KURT(DATA)".

CORRELATION

When a plot of data is made between one variable and another, with the presence of some amount of random variation in one of the variables, the resulting graph can look like one of those in Figure 27.6.

When the general shape of the data is upward-sloping to the right, the data are said to be positively correlated. When the shape is downward-sloping to the right, they are negatively correlated.

Mathematically, data are described by the statistic *r*, or Pearson's correlation coefficient. Pearson's correlation coefficient is used to measure the strength of a linear association between two variables, where the value $r = 1$ means a perfect positive correlation and the value $r = -1$ means a perfect negative correlation.

Use Excel's PEARSON() function. For example, for the data ranges named DATA1 and DATA2, use the Excel function expression "=PEARSON(DATA1, DATA2)".

LINEAR RELATIONSHIPS

Many physical relationships are known to be generally linear relationships. As an example, many elastic relationships are assumed to follow this rule. The relationship between the deflected length of a helical spring and the resulting force it produces is considered linear over a portion of its available deflection range. The current flowing in a purely resistive circuit is linearly related to voltage drop over a range of current values.

Departures from a linear relationship occur at the limits of such linear ranges. For example, compression springs are known to be nonlinear near the end limits of their available stroke (near their free length and also near their solid height). The relationship of current to voltage in an electrical circuit can become nonlinear when the value of resistance (or impedance) varies with the magnitude of current through it (for example, when heating effects change the magnitude of resistance).

A linear relationship is one where, if two or more pairs of data values (for example, two or more points in an x—y relationship) are plotted against one another, the resulting graph produces a straight line.

Two-Point Slope-Intercept Relationship in Linear Data Sets

When there are only two points, the line connecting them is simply a straight line. Such a linear graphical relationship can be described by a formula with two constants: one for the slope of the line and another for the intercept of the line with the y-axis.

This equation is the slope-intercept formula:

$$y = mx + b$$

where the formula for the slope is

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

The units of slope are a ratio: unit of y divided by the unit of x.

The formula for the intercept is:

$$b = y_1 - mx_1$$

The units of intercept are the same unit as y.

For example, in the x, y data pairs $x1 = 12.5$ mm, $y1 = 4.2$ N, $x2 = 22.1$ mm, $y2 = 8.4$ N, the slope is $m = 0.4$ N / mm, and the intercept is $b = –1.3$ N, rounded to one decimal place.

Use Excel's INTERCEPT() function. For example, for the data ranges named DATA1 and DATA2, use the Excel function expression "=INTERCEPT(DATA1, DATA2)".

Figure 27.8 illustrates a graph with slope and intercept.

Standard deviation $S_x = \sqrt{\frac{\sum x - \text{residuals}^2}{n-2}}$

and for the intercept (b):

$$b = \overline{y} - m\overline{x}$$

For example, in the following pair of data series of amperage measurements,

13, 14, 23, 23, 32, 33, 45, 99, and 105 A,

and induced amperage measurements,

34.60, 36.43, 59.23, 59.56, 82.03, 111.91, 240.50, and 255.19 A, respectively,

the slope of the best-fit line through the data pairs is 2.39, dimensionless, and the intercept is 3.66 A, rounded to two decimal places.

Use Excel's SLOPE() function. For example, for the data ranges named DATA1 and DATA2, use the Excel function expression "=SLOPE(DATA1, DATA2)".

Also, the intercept of the best-fit line through the data pairs is 3.66 A, rounded to two decimal places.

Use Excel's INTERCEPT() function. For example, for the data ranges named DATA1 and DATA2, use the Excel function expression "=INTERCEPT(DATA1, DATA2)".

ZERO AND SPAN RELATIONSHIPS

In most measuring and process control instruments, it is desired to have the displayed value, or value sent to a control final element (valve, motor, and so on), scaled such that a given lower output is used as a zero value, and the difference between an upper output and the lower zero value is termed as the *span*.

INTERPOLATION

Many numerical relationships are given in table (tabular) form. Often, acceptance criteria, standard values, and empirical data are transmitted in such format. The user of these data formats often needs to obtain intermediate values, by interpolation.

Interpolation is determining an intermediate output value in relation to one or more intermediate input values.

A related term, *extrapolation*, refers to estimating a value that is outside the range of given data, at a higher or lower value. Generally, extrapolation is to be discouraged unless specifically allowed by written guidance from the governing source of the data.

FORMATS OF TABULAR DATA

One-Way Tabulations. Tabular data are most often provided in input–output format. Each input data item value is related in a one-to-one relationship to an output value.

Two-Way Tabulations. Some data, such as in published steam tables, relate more than one input variable to an output variable. For example, in two-way tabulations, the first input is listed down a side column and the second across a row (usually at the top of the table), with the output variable listed as an array on a page. An example of this would be relating gas volume (output) to pressure and temperature (input variables).

Three-Way Tabulations. This is where three input variables are associated with an output variable. Extending the two-way thought, such tabulations have separate pages or tables for each value of the third input variable. An example of this is the *F*-distribution table.

LINEAR INTERPOLATION METHODS

Fortunately, much available data are either strictly linearly related or may be approximated satisfactorily with linear interpolation methods, over small known increments of the available tabular data. Depending on the type of data and the degree of accuracy required, the following interpolation methods are used.

One-Way, Two-Point Interpolation, Linear

One-way interpolation is used for graphical or tabular data, where the objective is to determine an intermediate output value in relation to a given intermediate input value.

A special case of one-way interpolation is where the input–output relationship is either known or assumed to be linear. This form of interpolation is most often used in practice, where the desired intermediate values are taken as linearly related.

The method for one-way, two-point linear interpolation involves locating known higher and lower input values, their related higher and lower output values, and use of the intermediate input value to compute the desired intermediate output value by the following formula:

$$y(x) = y_1 + \frac{x - x_1}{x_2 - x_1}(y_2 - y_1)$$

Here, $y(x)$ is the desired interpolated output value. x_1, x_2, y_1, and y_2 are the tabulated high and low input and output values, respectively.

For example, for the following source data voltage values, for points x and y,

x	y
1	8
2	1

the interpolated value of $y(x)$ at $x = 1.6$ V is $y(x) = 3.8$ V.

Two-Way, Two-Point Interpolation, Linear

For this requirement, the preceding formula is used incrementally, as follows. Using the formula and the intermediate value of the first input variable, determine two intermediate output values for the second input variable at both the high and low value of the first variable. Then repeat this process and, using the intermediate value of the second input variable, now having the values of the output variable at the intermediate value of the first input variable, compute the intermediate value of the output variable (which is the interpolated output value related to both intermediate input values).

For example, we have the following table of source data voltage values, for voltages at points w and x, related to a temperature, y °C, the interpolated value of $y(x)$ at $w_i = 77$ V, and $x_i = 86$ V is $y(w, x) =$ **0.507 V**.

Source data table for output temperature, y °C			
		x_1	x_2
		80	**90**
w_1	70	0.483	0.534 y
w_2	80	0.474	0.524 y

Three-Way Interpolation, Linear

Three-way interpolation is merely an extension of two-way interpolation, following the same logic. Obviously, the process becomes more and more repetitive and time-consuming.

In the CD-ROM the calculation is done automatically by providing the desired intermediate and tabulated input and output values from known tabular data.

For example, we have the following table of source data voltage values, for voltages at points w and x, related to a temperature, y °C, the interpolated value of z (y_i) at $w_i =$ **77 V**, $x_i =$ **86 *V*** and $y_i =$ **0.50**, is y (w, x, y) = **0.517 V**.

Source data tables for output temperature, y °C									
Y_1 = 0.25		x_1	x_2		**Y_1 = 0.75**		x_1	x_2	
		80	**90**				**80**	**90**	
w_1	70	0.483	0.534	z (y_1)	w_1	70	0.503	0.554	**z (y_2)**
w_2	80	0.474	0.524	z (y_1)	w_2	80	0.494	0.544	**z (y_2)**

INTERPOLATION METHODS FOR NONLINEAR DATA

Data that are slightly nonlinear, where higher-accuracy interpolated values are required, or substantially nonlinear data where known values between input and output values are more widely separated, require use of nonlinear interpolation methods. Generally, interpolation methods for such data are restricted to one-way (one input variable associated with each output variable) data, as the complexity and uncertainty can increase in nonlinear relationships.

TYPES OF DISTRIBUTIONS AND THEIR PROPERTIES

Normal (Gaussian) Probability Distribution

The *normal distribution*, or *gaussian distribution*, describes randomly occurring events and their frequency relative to the magnitude of their comparative measurement values. The density function of a normal probability distribution is bell shaped and symmetric about the mean.

The normal (gaussian) distribution is given by the normal probability density function formula in Figure 27.9 where x is normally distributed with a population mean μ and population variance σ^2.

$$f(x)=\frac{e^{-(x-\mu)^2/(2\sigma^2)}}{\sigma\sqrt{2}\pi}$$

Figure 27.9 Normal probability density function.

STANDARD NORMAL (GAUSSIAN) Z-DISTRIBUTION

The *standard normal (gaussian) distribution*, or *z-distribution*, is normalized to produce an area under the normal curve of 1.0 and place the peak of the curve at $z = 0$. This also means that the mean is 0 and the standard deviation is 1. This allows the convenient use of the z-statistic, also called the z-score, to locate the point where the area under the curve, at greater or lesser values than the z-statistic, is associated with the probability of occurrence of events having any of those z-statistic values. In other words, the z-score can tell you how many standard deviations from the mean each value lies.

The standard normal (gaussian) z-distribution is given by the formula in Figure 27.10.

$$z = \frac{x - \mu}{\sigma}$$

Figure 27.10 Calculating *z*-score.

Where x = individual value, μ = mean, σ = standard deviation.

To standardize a value from a normal distribution, convert the individual value into a z-score by:

1. Subtracting the mean from your individual value.
2. Dividing the difference by the standard deviation.

For example, if a weigh scale is assumed to have an approximately 100 % probability that the reading lies in the interval of ± 0.015 *N*, with a triangular distribution, then the standard uncertainty u_i is 0.0061 *N*. These types of uncertainty contributors are often used for noise and vibration.

Uncertainty, U-Shaped Distribution

U-shaped distributions generally apply to distributions where the frequency of occurrence of the measurement is lowest at the mean of the distribution, increasing in some manner for values above or below the distribution mean.

For an assumed U-shaped distribution, the Type B standard uncertainty u_j, where 2*a* is the range including approximately 100 % probability that the value lies in the interval—*a* to +*a*, is given by the following formula:

$$u_i = a/\sqrt{2}$$

For example, if a thermometer is assumed to have an approximately 100% probability that the reading lies in the interval of ± 0.1 K, with a U-shaped distribution, the standard uncertainty u_j then is 0.07 K. This distribution is used when referencing cyclic influences, such as temperature variation.

Relative Uncertainty

Relative uncertainty, U_r, is a dimensionless value that expresses a unit's uncertainty independent of the unit's size. It is the measurement uncertainty, *U*, divided by the magnitude of the unit, *y*, given by the following formula:

$$U_r = u/y$$

For instance, if an ammeter measures (y = 0.75 A), and the uncertainty, *u*, of the measurement is known to be 0.005 A, then the relative uncertainty U_r is 0.00667, dimensionless.

AUTOCORRELATION

One important test of the assumption that source data are normally distributed is to determine the *autocorrelation coefficient* for the data set, a value between 0 and 1. If the autocorrelation coefficient is near zero, the data set may be more confidently assumed to be normally distributed. If it is larger, this indicates the presence of cyclic variation in the data, and the presence of a sequence- or time-based factor in the data. The autocorrelation function can be used for the following two purposes:

1. To detect nonrandomness in data
2. To identify an appropriate time series model if the data are not random

The general formula for calculating the autocorrelation for lag_k, denoted r_k, given measurement, $Y_1, Y_2, \ldots, Y_N$ *at time* $X_1, X_2, \ldots, X_N$ is seen in Figure 27.17.